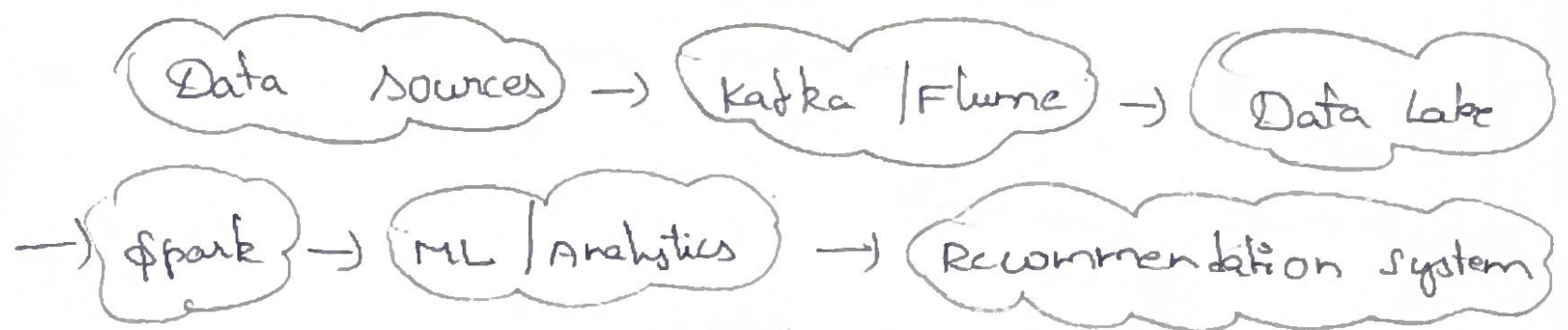


1 Unified Big Data Architecture for Retail

A retail organization can use a centralized data lake to combine online, offline, CRM and transaction data.

Architecture:



=> Real-time data is processed using Spark streaming.

=> Customer profiles are created by integrating all channels.

=> ML models provide personalized recommendations

=> Cloud storage provides scalability.

2. Distributed computing for scientific Data

Distributed computing divides large scientific datasets into smaller tasks processed by multiple machines.

=> Hadoop MapReduce / Spark enables parallel processing.

=> Reduces computation time.

=> Provides scalability by adding nodes

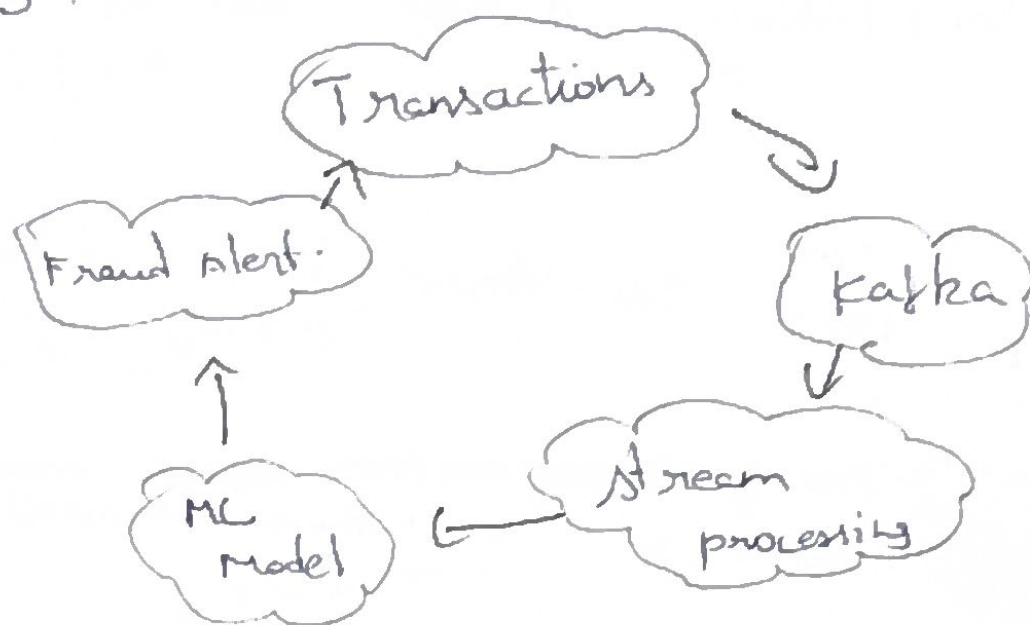
=> Fault tolerance is achieved through data replication.

=> It is more efficient than a single centralized system for huge datasets

3. Big Data Fraud Detection Framework:

A payment platform can use kafka + spark streaming + machine Learning.

Flow.



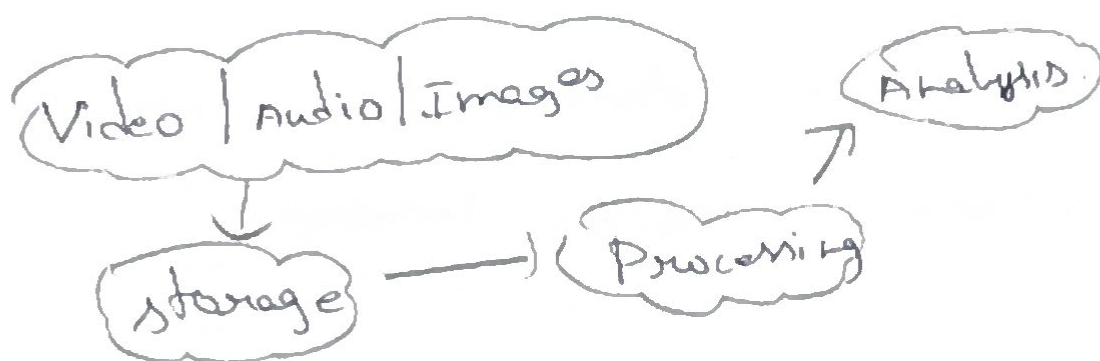
⇒ Spark streaming detects suspicious transactions immediately.

⇒ ML identifies unusual behaviour and transaction patterns.

⇒ Historical data improves model accuracy.

⇒ Alerts can automatically block or verify suspicious payments.

4. Multimedia Big Data storage.



=> Use cloud/object storage such as distributed storage.

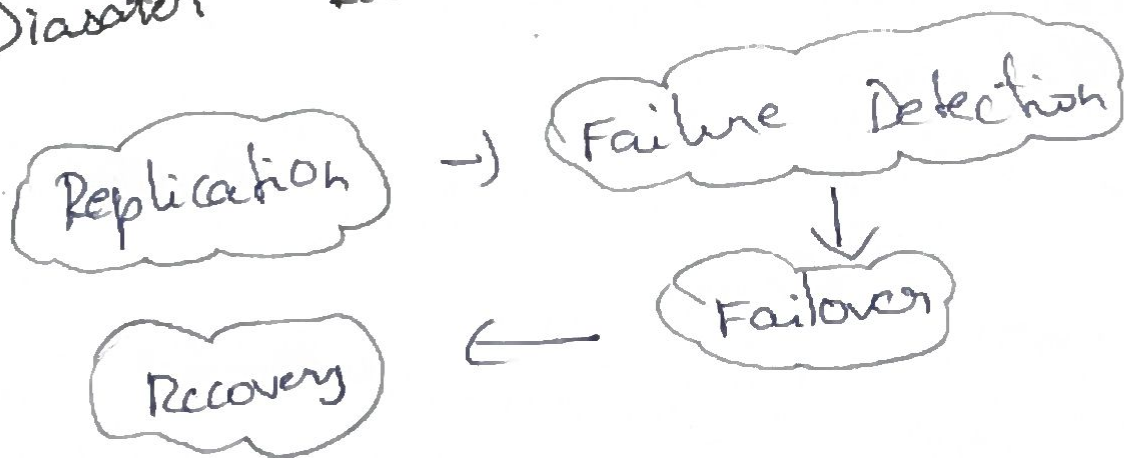
=> Hadoop HDFS can store large multimedia datasets.

=> Use compression to reduce storage requirements.

=> Metadata and indexing improve searching.

=> Distributed processing helps analyze multimedia efficiently.

5. Disaster Recovery and Business Continuity.



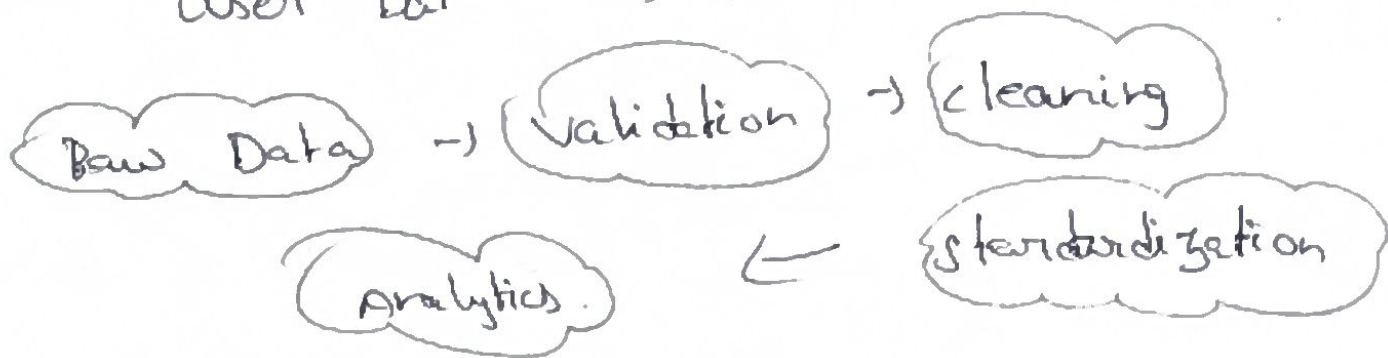
- => Replication - maintains duplicate data
Copies
- => Redundancy - avoids single points of failure
- => Automated failover - switches to backup servers automatically.
- => Regular backups - enable data recovery.
- => Multi-region / Global deployment improves availability.

Thus, services can continue even when hardware or software fails.

b. Big Data Quality and Governance.

Common data quality problems include missing, duplicate, inconsistent and incorrect data.

User Data → Processing → ML Model



→ Data validation and cleansing.

=> Deduplication

=> Handling missing values

=> Standardizing data formats

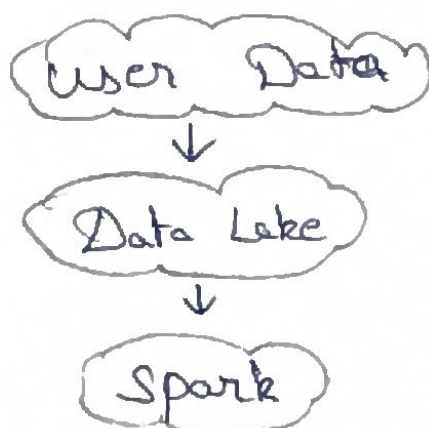
=> Data governance policies.

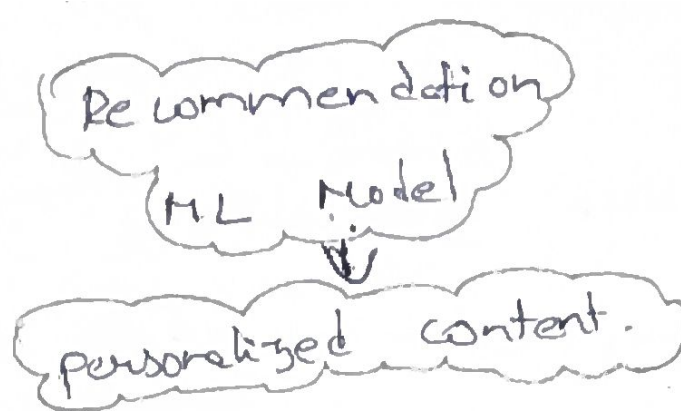
=> Access control and data ownership

Good data quality improves the reliability of analytics.

7. Scalable Recommendation System.

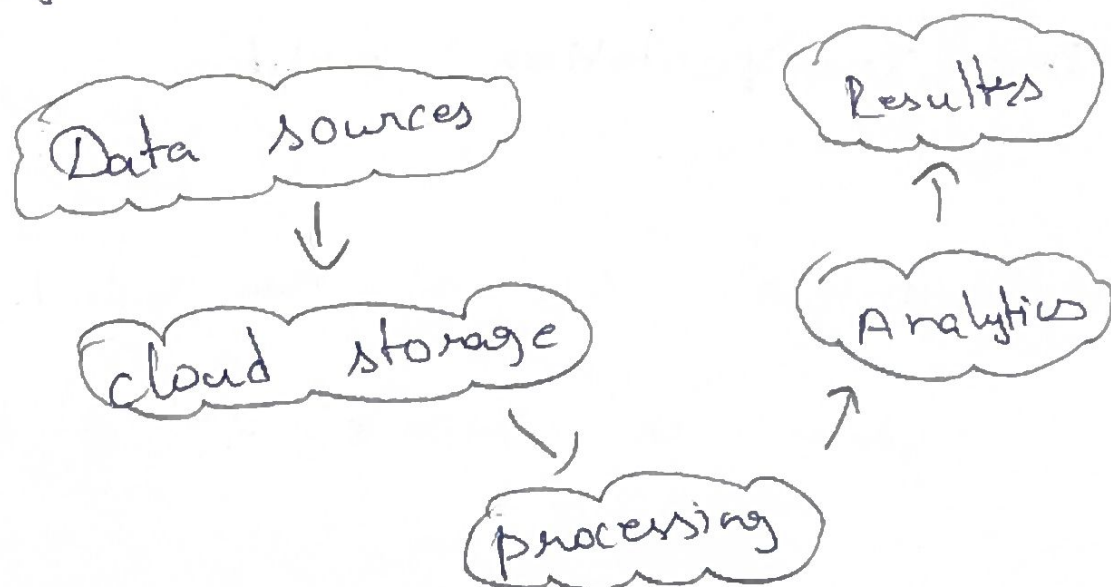
A streaming service can collect watch history, ratings, searches, and user behaviour





- =) Distributed processing handles millions of users
- =) ML identifies user preferences.
- =) Real-time behaviour updates recommendations.
- =) Personalized recommendations increase accuracy and user engagement.

8. cloud computing in Big Data.
cloud computing provides flexible infrastructure for big data applications.



Advantages:

- =) High Scalability
- =) Pay-as-you-use cost model
- =) Easy resource expansion
- =) Flexible storage and computing
- =) High availability.

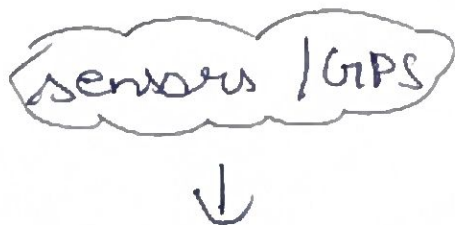
Limitations:

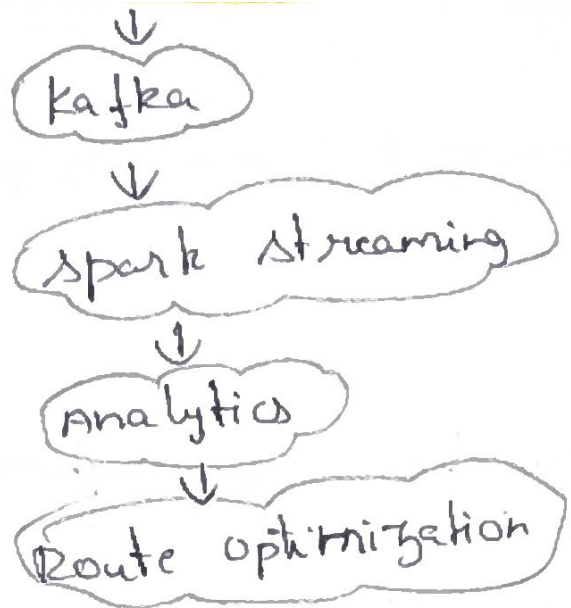
- =) Security and privacy concerns
- =) Internet dependency
- =) Possible vendor lock-in
- =) Large-scale data transfer can be costly.

9. Real-Time Transportation Analytics.

Transportation systems can collect GPS, traffic, weather and vehicle sensor data.

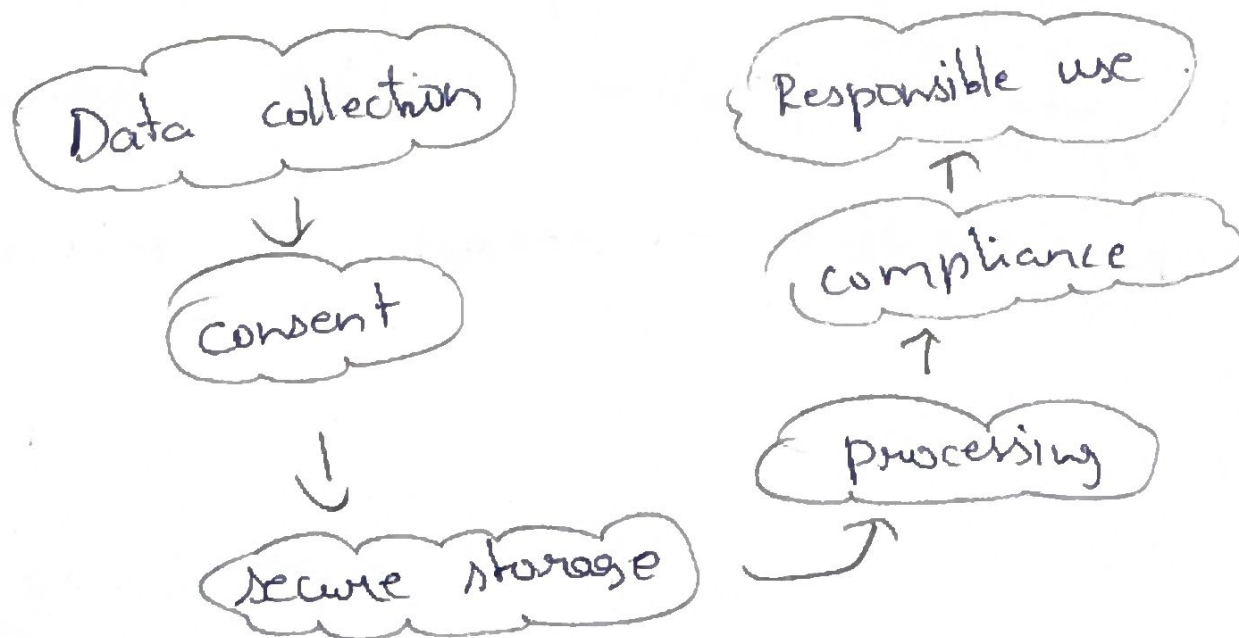
Flow:





- => Detects traffic conditions in real time
- => Finds optimal routes
- => Reduces fuel consumption.
- => Predicts delays
- => Improves fleet management and operational decisions.

10. Ethical and Legal challenges in Big Data



Privacy: unauthorized use of personal data

Security: data breaches and cyberattacks

Consent: users should know how data is collected.

Bias: analytics / AI may produce unfair results.

Legal compliance: organizations must follow data protection laws.

Policies: use consent, encryption, access control, data minimization, audits, transparency and proper data - retention policies